Math 54 — Week 10

Solutions to the problem set

P1. **Idea.** Subtract from $\mathbf{x}_2$ its projection onto $\mathbf{v}_1 = \mathbf{x}_1$, clear fractions, then normalize.

1. $\mathbf{v}_1 = (1,2,2)$, with $\mathbf{v}_1 \cdot \mathbf{v}_1 = 9$, and $\mathbf{x}_2 \cdot \mathbf{v}_1 = 4+2+2 = 8$. So
$$\mathbf{v}_2 = (4,1,1) - \frac{8}{9}(1,2,2) = \Big(\frac{28}{9},\ -\frac{7}{9},\ -\frac{7}{9}\Big) = \frac{7}{9}(4,-1,-1).$$
2. Use $(4,-1,-1)$ instead. Check: $(1,2,2)\cdot(4,-1,-1) = 4-2-2 = 0$.
3. The lengths are $\|(1,2,2)\| = 3$ and $\|(4,-1,-1)\| = \sqrt{18} = 3\sqrt{2}$, so an orthonormal basis is
$$\frac{1}{3}(1,2,2), \qquad \frac{1}{3\sqrt{2}}(4,-1,-1).$$

P2. **Idea.** Gram–Schmidt on the three vectors in the order given, clearing fractions after each step.

1. $\mathbf{v}_1 = (1,1,1,1)$, with $\mathbf{v}_1 \cdot \mathbf{v}_1 = 4$.
2. $\mathbf{x}_2 \cdot \mathbf{v}_1 = 3$, so $\mathbf{x}_2 - \frac{3}{4}\mathbf{v}_1 = (\frac{1}{4},\frac{1}{4},\frac{1}{4},-\frac{3}{4})$. Use $\mathbf{v}_2 = (1,1,1,-3)$, with $\mathbf{v}_2 \cdot \mathbf{v}_2 = 12$.
3. $\mathbf{x}_3 \cdot \mathbf{v}_1 = 2$ and $\mathbf{x}_3 \cdot \mathbf{v}_2 = 2$, so
$$\mathbf{x}_3 - \frac{2}{4}\mathbf{v}_1 - \frac{2}{12}\mathbf{v}_2 = (1,1,0,0) - \Big(\frac{1}{2},\frac{1}{2},\frac{1}{2},\frac{1}{2}\Big) - \Big(\frac{1}{6},\frac{1}{6},\frac{1}{6},-\frac{1}{2}\Big) = \Big(\frac{1}{3},\frac{1}{3},-\frac{2}{3},0\Big).$$
Use $\mathbf{v}_3 = (1,1,-2,0)$.
4. Check: $\mathbf{v}_3 \cdot \mathbf{v}_1 = 1+1-2 = 0$ and $\mathbf{v}_3 \cdot \mathbf{v}_2 = 1+1-2+0 = 0$. An orthogonal basis is $(1,1,1,1)$, $(1,1,1,-3)$, $(1,1,-2,0)$.

P3. **Idea.** Q comes from P1; then $R = Q^{\mathsf{T}}A$.

1. From P1, $\mathbf{q}_1 = \frac{1}{3}(1,2,2)$ and $\mathbf{q}_2 = \frac{1}{3\sqrt{2}}(4,-1,-1)$.
2. $r_{11} = \mathbf{q}_1 \cdot \mathbf{x}_1 = \frac{9}{3} = 3$, $r_{12} = \mathbf{q}_1 \cdot \mathbf{x}_2 = \frac{8}{3}$, and
$$r_{22} = \mathbf{q}_2 \cdot \mathbf{x}_2 = \frac{16-1-1}{3\sqrt{2}} = \frac{14}{3\sqrt{2}} = \frac{7\sqrt{2}}{3}.$$
3. So
$$Q = \begin{bmatrix} 1/3 & 4/(3\sqrt{2}) \\ 2/3 & -1/(3\sqrt{2}) \\ 2/3 & -1/(3\sqrt{2}) \end{bmatrix}, \qquad R = \begin{bmatrix} 3 & 8/3 \\ 0 & 7\sqrt{2}/3 \end{bmatrix}.$$
4. Check column 2:
$$\frac{8}{3}\,\mathbf{q}_1 + \frac{7\sqrt{2}}{3}\,\mathbf{q}_2 = \frac{8}{9}(1,2,2) + \frac{7}{9}(4,-1,-1) = (4,1,1).$$
The diagonal entries 3 and $\frac{7\sqrt{2}}{3}$ are the lengths of $\mathbf{v}_1$ and of $\mathbf{v}_2 = \frac{7}{9}(4,-1,-1)$.

P4. **Idea.** The columns are independent, so the normal equations have one solution.

1. The normal equations:
$$A^{\mathsf{T}}A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \qquad A^{\mathsf{T}}\mathbf{b} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$
2. The determinant is 3, so
$$\hat{\mathbf{x}} = \frac{1}{3}\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1/3 \\ 1/3 \end{bmatrix}.$$
3. $A\hat{\mathbf{x}} = (\frac{1}{3}, \frac{1}{3}, \frac{2}{3})$ and $\mathbf{b} - A\hat{\mathbf{x}} = (\frac{2}{3}, \frac{2}{3}, -\frac{2}{3})$. Check: the residual is orthogonal to both columns, $(1,0,1)\cdot(\frac{2}{3}, \frac{2}{3}, -\frac{2}{3}) = 0$ and $(0,1,1)\cdot(\frac{2}{3}, \frac{2}{3}, -\frac{2}{3}) = 0$.
4. The least-squares error is
$$\|\mathbf{b} - A\hat{\mathbf{x}}\| = \frac{2}{3}\sqrt{3} = \frac{2\sqrt{3}}{3}.$$

P5. **Idea.** Solve $R\hat{\mathbf{x}} = Q^{\mathsf{T}}\mathbf{b}$ from the bottom row up.

1. Since $\mathbf{b} = (1,0,0)$, the entries of $Q^{\mathsf{T}}\mathbf{b}$ are the first entries of $\mathbf{q}_1$ and $\mathbf{q}_2$:
$$\mathbf{q}_1 \cdot \mathbf{b} = \frac{1}{3}, \qquad \mathbf{q}_2 \cdot \mathbf{b} = \frac{4}{3\sqrt{2}} = \frac{2\sqrt{2}}{3}.$$
2. The second row of $R\hat{\mathbf{x}} = Q^{\mathsf{T}}\mathbf{b}$ is
$$\frac{7\sqrt{2}}{3}\,\hat{x}_2 = \frac{2\sqrt{2}}{3},$$
so $\hat{x}_2 = \frac{2}{7}$.
3. The first row gives $3\hat{x}_1 + \frac{8}{3}\cdot\frac{2}{7} = \frac{1}{3}$, so
$$3\hat{x}_1 = \frac{1}{3} - \frac{16}{21} = -\frac{3}{7}, \qquad \hat{x}_1 = -\frac{1}{7}.$$
4. Check with the normal equations: $A^{\mathsf{T}}A$ has rows $(9,8)$, $(8,18)$ and $A^{\mathsf{T}}\mathbf{b} = (1,4)$, and
$$9\Big(-\frac{1}{7}\Big) + 8\cdot\frac{2}{7} = 1, \qquad 8\Big(-\frac{1}{7}\Big) + 18\cdot\frac{2}{7} = 4.$$

P6. **Idea.** Orthonormal columns make $A^{\mathsf{T}}A = I$, so the normal equations are already solved.

1. The columns are orthonormal, so $A^{\mathsf{T}}A = I$ (W4). They are also independent (W3), so the least-squares solution is unique (Lay Thm. 14).
2. The normal equations $A^{\mathsf{T}}A\mathbf{x} = A^{\mathsf{T}}\mathbf{b}$ become $\mathbf{x} = A^{\mathsf{T}}\mathbf{b}$, so $\hat{\mathbf{x}} = A^{\mathsf{T}}\mathbf{b}$.
3. $A\hat{\mathbf{x}}$ is the projection of $\mathbf{b}$ onto $\operatorname{Col} A$, so $\operatorname{proj}_{\operatorname{Col} A}\mathbf{b} = AA^{\mathsf{T}}\mathbf{b}$. (This is Week 9, Thm. 10 again.)

P7. **Idea.** Work with the value vectors at $0, 1, 2$; the inner product is their dot product.

1. The value vectors are $p = 1 + t \mapsto (1, 2, 3)$ and $q = t^2 \mapsto (0, 1, 4)$. So
$$\langle p, q\rangle = 0 + 2 + 12 = 14, \qquad \|p\| = \sqrt{1 + 4 + 9} = \sqrt{14}, \qquad \|q\| = \sqrt{0 + 1 + 16} = \sqrt{17}.$$
2. Gram–Schmidt: $p_0 = 1 \mapsto (1, 1, 1)$, with $\langle 1, 1\rangle = 3$. Then $\langle t, 1\rangle = 0 + 1 + 2 = 3$, so $p_1 = t - 1 \mapsto (-1, 0, 1)$, with $\langle p_1, p_1\rangle = 2$.
3. $\langle t^2, 1\rangle = 5$ and $\langle t^2, t - 1\rangle = 0 \cdot (-1) + 1 \cdot 0 + 4 \cdot 1 = 4$, so
$$p_2 = t^2 - \frac{5}{3} \cdot 1 - \frac{4}{2}(t - 1) = t^2 - 2t + \frac{1}{3}.$$
4. Check: $p_2 \mapsto (\frac{1}{3}, -\frac{2}{3}, \frac{1}{3})$, which has dot product 0 with $(1, 1, 1)$ and with $(-1, 0, 1)$. An orthogonal basis is 1, $t - 1$, $t^2 - 2t + \frac{1}{3}$.

P8. **Idea.** Choose $\mathbf{u}$ so that $\langle \mathbf{u}, \mathbf{v}\rangle$ is the sum or integral on the left, then square Cauchy–Schwarz.

1. (a) In $\mathbb{R}^3$ take $\mathbf{u} = (1, 1, 1)$ and $\mathbf{v} = (a, b, c)$. Then $\mathbf{u} \cdot \mathbf{v} = a + b + c$, $\|\mathbf{u}\| = \sqrt{3}$ and $\|\mathbf{v}\| = \sqrt{a^2 + b^2 + c^2}$. Squaring both sides of the Cauchy–Schwarz inequality $|a + b + c| \le \sqrt{3}\sqrt{a^2 + b^2 + c^2}$ gives (a).
2. (b) In $C[0, 1]$ with the integral inner product take $\mathbf{u} = 1$ and $\mathbf{v} = f$. Then $\langle 1, f\rangle = \int_0^1 f\,dt$ and $\|1\| = 1$. Squaring both sides of the Cauchy–Schwarz inequality
$$\left|\int_0^1 f(t)\,dt\right| \le 1 \cdot \left(\int_0^1 f(t)^2\,dt\right)^{1/2}$$
gives (b).